

1 Data Structure of Components

Production resources and tools are stored in the table mlst_hy with resource type "MAT". In the sections below, the different fields of a (material) component are described. The actual order of fields in the editing dialogs can deviate from the order used here. Not all fields that are listed here are displayed on the client or can be edited.

MES order number / MES work plan number

The component is assigned to the operation identified in this field. The field shows the combined order and operation number or work plan and operation number of the operation.

Material

Enter the material number of the material component.

Designation

You can enter the name of the material.

Comment 1 / Comment 2

These are comment fields.

BOM item

The BOM lists the different components of a product. The components are referred to as items. The number entered here specifies the position where the item is listed in the BOM. It is therefore possible that the BOM includes one material number several times. Using the correct BOM item, the data collection can still be uniquely assigned.



Note: when using the MPL, the BOM item must be **unique** for an operation. Each component must have a unique BOM item if several components are used in one operation. Two components must not have the same BOM item.

For the coil-based solution "RF", the BOM item specifies the position of the component in the layer structure.

BOM level

A component can also have several levels. If applicable and known, enter the BOM level here.



If you log on input batches via material management (MPL/TRT), you can only log on components of BOM level 0.

If you enter a BOM level > 1, the system automatically sets the component type (see second next field) to "I" (info component).

Material type

Material type of the material component. The material type controls the material-specific processing in the system.

Unless defined otherwise for a specific project, assign the material type SYSTEM here.



The material type must be available in the system (see configuration of [Material types](#)). If no material type has been entered, the system tries to identify the material component (requirement: the [assignment of material to material type](#) has been made). If the system cannot identify the material type, the system uses the material type "SYSTEM".



For info components (material type "I"), we recommend to define and assign a separate material type (e.g. INFO).

Component type

Possible values:

- | | |
|----------|--|
| M | Material component (default)
You usually enter "M" here. Other component types can be relevant for material management ("MPL"). |
| I | Info component
You can display info components in the bill of materials (BOM), but you need not log them on or off. |
| T | Carrier material (coil-based production)
You can log on a maximum of one input batch as carrier material (T) or added material (Z) to the machine. |
| A | Scrap/waste material (coil-based production) |
| Z | Added material as alternative for the carrier (coil-based production)
You can log on a maximum of one input batch as carrier material (T) or added material (Z) to the machine. |

Consumption type

The following collection options are available for material components. The definition of the different options and their use depend on the functions used.

N = None

This option defines that no consumption is collected for the material component. The material component is only displayed here.

The so-called info components (see above: component type) must be set to this option.

L = Retrograde/with batch reference (MPL/TRT, MPL-RF)

If this option is used, the material component is logged on and off as batch. The consumption calculation for this material component (retrograde, at input batch logoff) depends on the configuration of the [material type](#) the material component is assigned to.

D = Discrete

This option is relevant for *discrete consumption recording* (AIP-DVE). This type of material consumption recording requires a configuration that can be part of a customization at the customer's.

Use the option "L" if Material and Production Logistic (MPL) or Tracking & Tracing (TRT) is used.

For this component, the system calculates consumption using the quantity produced last and suggests this calculated quantity in a posting dialog. The consumption is posted for the component and a material movement is generated (goods issue from production). This material movement can be uploaded to the higher-level ERP system.

Replaceable

If this identifier (=J) is set, you can use a different material than the material planned for this component. You can only use a material of the same material type.

For the user on the shop floor client (MPL/TRT) a message is displayed. The user selects and logs on the relevant component.

Change necessary / Requirement to change output batch

With this option, an input batch change for a batch of this material forces an output batch change. The setting that is allowed for this option depends on the relevant component type (see above):

Component type	Allowed settings
M	☐ or ☒ possible
T, Z	Only ☒ allowed
I, A	Only ☐ allowed

Input quantity

Input quantity of the component per unit in primary quantity that is planned for the operation.

Unit

Quantity unit of input quantity

Input quantity in percent / Upper tolerance limit / Lower tolerance limit

Default: 0; should only be modified after consultation with MPDV.

Required quantity

Total quantity of the component that is planned for the operation.

The system calculates the required quantity when the display is called. The following formula is used for calculation :

Required quantity = input quantity of component x target quantity of OP in primary quantity unit

The required quantity is only displayed in the table and in the detail panel.

Resource type

Reserved. Not used.

UOM Spec. mass per unit area

Reserved. Not used.

Spec. mass per unit area

Reserved. Not used.

Planned article

Reserved. Not used.

Backflush

Reserved. Not used.

Required quantity (PPS)

The field *Required quantity* shows the total quantity required that is transferred from the ERP/PPS system. That is the quantity of the component that is required to produce the target quantity of the operation.

Consumption (total)

The column *Consumption (total)* shows the total consumption that has been posted for the relevant component. In this context, it does not matter whether the component is subject to batch management or discrete.

Upper-level component: BOM item / BOM level

Reserved. Not used.

Modified by / Modified on

Editor and date and time of the last change

User fields

You can define and use user fields for specific projects.